

# The Interaction Zone: State Financial Literacy and the Cross-Section of the Momentum Anomaly

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## **Abstract**

Financial literacy varies widely across U.S. states, and less literate investors are known to make worse portfolio choices. If those choices move prices, state literacy should moderate the return anomalies driven by retail mispricing. In aggregate the signal is weak, but I show it is not absent so much as hidden by aggregation. On the stable-headquarters subsample of CRSP common stock from 2009 to 2023, the literacy moderation of the momentum and idiosyncratic-volatility interaction concentrates in firms with mid-level institutional ownership: the segment with enough retail demand to move prices and

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enough institutional activity to register that demand in returns across states. Firms held mostly by retail and firms dominated by institutions show no moderation. Restricting the sample to larger firms moves the moderation to the low-ownership tercile, and the cross-state literacy gradient moves with it. This migration with size composition is the paper’s main piece of evidence: it locates where the mechanism operates without relying on an instrument, and it bounds the result to firms whose headquarters, and hence local investor base, stay put.

## 1 Introduction

Lusardi and Mitchell [2014] and a large literature document that financially literate households make better portfolio decisions: they diversify more, panic less, time the market less, and pay more attention to fees and volatility-adjusted risk. The variation across U.S. states is substantial. Roughly 28 percent of adults answer three of five basic literacy questions correctly in high-literacy states like Minnesota and New Hampshire, against around 21 percent in Mississippi and West Virginia. The local-bias literature supplies the second piece: retail investors hold portfolios tilted toward locally headquartered firms [Coval and Moskowitz, 1999, Ivković and Weisbenner, 2005], and this local demand moves prices [Hong et al., 2008, Bernile et al., 2015]. Together they yield a clean hypothesis. Where retail investors are less literate, they should trade more like the textbook noise trader, so state literacy should moderate the momentum-IV interaction, a retail-friendly anomaly

paired with the standard limits-to-arbitrage measure, with the effect weaker in literate states.

In aggregate, the data do not deliver. Regressing the three-way interaction on the full panel of CRSP-listed common stock returns a modest negative coefficient with marginal significance, and the magnitude all but vanishes once headquarters-state assignment is corrected for the 14 percent of firm-months whose static Compustat snapshot does not match the then-current 10-K filing. But the aggregate test pools firms with very different ownership structures, and it pools firms whose local-investor base is well-defined (stable headquarters) with firms whose base is in transition (relocators). So the aggregate null says little about whether a literacy channel exists. It says only that the channel does not operate uniformly across the cross-section.

The contribution of this paper is to ask a sharper question: where in the cross-section does the literacy mechanism become visible, under what conditions on firm ownership and size, and what does the answer say about the microfoundation? On the stable-headquarters subsample of CRSP common stock from 2009 to 2023, the literacy moderation concentrates in firms whose institutional ownership sits in the middle of the within-stock IO distribution, the mid-IO tercile, with mean ownership share around fifty percent. Firms with predominantly retail ownership (low IO) show essentially no moderation, and firms with predominantly institutional ownership (high IO) show none either. The mid-tercile coefficient survives once firm size enters as a competing moderator, so the finding is not a size effect relabeled as owner-

ship, even though mid-IO firms are predominantly mid-cap. Within mid-IO, the coefficient maps to a cross-state literacy gradient in the predicted direction: higher-literacy states show weaker anomaly slopes, and the across-state correlation between per-state coefficients and state literacy is clear.

The interpretation the data point to is an *interaction zone*. The literacy channel needs two ingredients to leave a price footprint: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into return differences detectable across states. Where neither is present (micro-cap firms held mostly by retail), retail mispricing is real but accumulates uniformly and does not load on cross-state literacy variation. Where institutions dominate (large firms with high IO), arbitrage absorbs retail demand and neutralizes the local-state channel. The interaction zone is the set of firms with mixed ownership at meaningful market size, where both ingredients are present. In the full stable-HQ panel it falls predominantly in mid-IO; dropping the bottom 50 percent of firms by market capitalization shifts the moderation to low-IO, and the cross-state gradient shifts with it. It is the same economic population, relocated to a different tercile of the conditional IO distribution.

This migration is the paper’s main piece of mechanism evidence, the signature an interaction-zone reading predicts and the IO-tercile result alone does not deliver. The high-IO tercile shows no cross-state gradient under any sample restriction, the null the reading predicts. A decomposition by idiosyncratic-volatility decile within mid-IO shows none of the strengthening-

at-high-IV pattern the textbook limits-to-arbitrage story would require, which points the mechanism toward size and ownership conditioning rather than arbitrage cost.

Two facts bound the result. The continuous-margin quadratic-IO U-shape clears five percent on a quadratic form only, not under log or kink alternatives, so the functional-form-free claim is the discrete-tercile contrast. And under point-in-time literacy reassignment on the full panel the mid-IO coefficient shrinks sharply, so the characterization is a statement about firms with stable headquarters rather than the corrected full panel. An instrument built from state K-12 personal-finance graduation requirements is not available here, because mandate states are systematically less literate (states adopted mandates in response to low adult literacy); the OLS estimate is the preferred inferential statement, and the migration supplies the predicted-signature confirmation an instrument would otherwise provide.

The closest prior work is Korniotis and Kumar [2013], who show that state-level business cycles predict returns of locally headquartered firms. This paper differs on three margins: the state variable is NFCS literacy rather than the business cycle, the object is a three-way moderation of an anomaly interaction rather than a level prediction, and the discriminating margin is a within-stock IO contrast the business-cycle channel does not predict. A business-cycle control block attenuates the literacy three-way by 20 to 23 percent, so the two state moderators are distinct but partly overlapping. Korniotis et al. [2020] use brokerage-demographic sophistication proxies but

neither NFCS literacy nor the momentum-IV interaction, and Hillert et al. [2014] is the closest precedent for a state-level moderator of momentum, using media coverage with state individualism as the cross-sectional moderator rather than literacy.

**Roadmap.** Section 2 describes the data, the NFCS state-literacy panel, and the stable-headquarters subsample. Section 3 states the specification and identifying assumptions. Section 4 reports the mid-IO concentration, the size-controlled extension, the migration, and the cross-state gradient. Section 5 develops the interaction-zone characterization. Section 6 reports the robustness checks. Section 7 discusses interpretation and limitations, and Section 8 concludes.

## 2 Data and Sample Construction

### 2.1 Sample

The sample is the cross-section of CRSP-listed common stock (share codes 10 and 11; exchange codes 1, 2, 3, covering NYSE, NYSE American, and NASDAQ) from January 2009 through December 2023. The full panel contains 623,896 firm-month observations, 7,447 unique firms, 51 headquarters states (the fifty states plus the District of Columbia), and 180 calendar months. I z-score all focal variables within each calendar month, so the residual variation the regressions exploit is cross-sectional and free of time-aggregate move-

ments. State financial literacy is measured from the NFCS Big Five literacy questions, fielded by the FINRA Investor Education Foundation in waves 2009, 2012, 2015, 2018, and 2021, and forward-filled to monthly frequency between waves.

## 2.2 The stable-headquarters subsample

The local-bias channel transmits through community ties between a firm and its headquarters-state retail base, and those ties take time to form and decay, so for firms that relocate within sample the effective local-investor population is ambiguous. I drop firms with a confirmed in-sample EDGAR-based relocation, using the union of two flags. The first, `hdr_changed`, fires when the 10-K-header business-address `STATE` changes across a firm’s in-sample 10-Ks (954 firms); the second, `sec_disagree`, fires when the most-recent 10-K-header state disagrees with the static Compustat snapshot and the firm has at least two parsed headers (524 additional firms). The union identifies 1,366 relocators (16.8 percent of firm-months), and the top first-to-last state pairs are consistent with the relocation mechanism of Pirinsky and Wang [2006]. The headline patterns are estimated on the resulting stable-headquarters subsample (519,090 firm-months, 6,081 firms) under static-snapshot literacy; the point-in-time reassignment on the full panel is the correctness check (Section 6.1). Construction and PIT-coverage detail are in IA A and IA B.

## 2.3 Variable definitions

The dependent variable is the monthly stock return `ret` from CRSP. Momentum is `mom_12_2`, the standard Jegadeesh and Titman [1993] cumulative return from twelve months before to two months before the observation month, skipping the most recent month. Idiosyncratic volatility (IV) is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. State financial literacy is the NFCS Big Five share-correct score, z-scored within month and denoted `literacy_z`.<sup>1</sup> Each cross-product (the two-way terms  $\text{mom} \times \text{IV}$ ,  $\text{mom} \times \text{literacy\_z}$ ,  $\text{IV} \times \text{literacy\_z}$ , and the three-way  $\text{mom} \times \text{IV} \times \text{literacy\_z}$ ) is z-scored within month. The Fama-French 12-industry block and the Korniotis and Kumar [2013] business-cycle block enter robustness specifications.

## 2.4 Institutional ownership from the Thomson Reuters

### `s34 13F` panel

I build a firm-quarter institutional-ownership (IO) panel from the WRDS Thomson Reuters s34 13F holdings file for 2009Q1 through 2023Q4 (222,180 firm-quarter observations, 7,907 distinct permnos). The numerator follows the literature-standard convention: amendment-deduplicated filings (latest filing per manager-security-report-date triple) and the sole-plus-shared investment-discretion measure rather than the raw `shares` field. This brings the residual

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<sup>1</sup>Correct-answer keys M6=1, M7=3, M8=2, M9=2, M10=1, verified from the FINRA Foundation NFCS State-by-State Codebook (2021); full construction detail in Appendix A.

above-one rate to 0.36 percent, against 6.49 percent under raw `shares`, and I cap it at one. Full construction detail is in IA P.

The persistent IO measure is each firm’s time-mean institutional-ownership share across its available Thomson quarters; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier (a strictly conservative no-look-ahead merge). On the stable-headquarters subsample, persistent IO covers 518,962 firm-months and time-varying IO covers 493,889 firm-months.

## 2.5 Descriptive statistics

Table 1 reports the cross-sectional distribution of the focal variables.

# 3 Identification

The introduction posed two questions: where in the cross-section the literacy mechanism becomes visible, and what microfoundation that location identifies. This section describes the design that answers them.

## 3.1 What the research question requires

Testing whether state literacy moderates the momentum-IV anomaly, more strongly in literate states, means measuring how the within-stock relationship between returns and  $\text{mom} \times \text{IV}$  varies with the headquarters-state literacy of each stock. Three features of the data dictate the design. First, literacy is

Table 1: Descriptive statistics and state literacy rankings  
*Panel A: Distribution of focal variables (full panel)*

| Variable                           | Mean  | SD    | p05    | p50   | p95   |
|------------------------------------|-------|-------|--------|-------|-------|
| Monthly return                     | 0.013 | 0.192 | −0.220 | 0.005 | 0.253 |
| Momentum ( <code>mom_12_2</code> ) | 0.107 | 0.828 | −0.678 | 0.034 | 1.021 |
| Idiosyncratic volatility           | 0.027 | 0.022 | 0.007  | 0.020 | 0.070 |
| Market equity (\$ billions)        | 7.04  | 42.1  | 0.017  | 0.596 | 26.19 |
| NFCS Big Five share correct        | 0.244 | 0.028 | 0.200  | 0.247 | 0.286 |

*Panel B: Panels and merge conventions*

| Panel                                                    | Firm-months | Firms |
|----------------------------------------------------------|-------------|-------|
| Full panel (static-snapshot HQ, baseline literacy merge) | 623,896     | 7,447 |
| Full panel, PIT literacy reassignment                    | 623,814     | 7,447 |
| Stable-HQ subsample (no in-sample EDGAR relocation)      | 519,090     | 6,081 |
| with persistent IO measure                               | 518,962     | 6,018 |
| with time-varying IO measure                             | 493,889     | 5,877 |

*Panel C: State literacy rankings and NFCS wave summary*

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Highest-literacy states                           | MN, HI, NH, DC, CT                |
| Lowest-literacy states                            | MS, OH, AR, TX, WV                |
| NFCS wave means (2009, 2012, 2015, 2018, 2021)    | 0.258, 0.254, 0.255, 0.229, 0.222 |
| Median within-state rank correlation across waves | 0.71 (Spearman)                   |

*Notes.* Idiosyncratic volatility is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. Market equity is price times shares outstanding from CRSP, in billions of dollars. The NFCS Big Five share-correct score is the state-wave share of respondents answering the FINRA Big Five financial-literacy questions correctly. The within-state rank correlation across the five NFCS waves is the median Spearman correlation of state-wave ranks across consecutive waves (a measurement-stability statistic for the state literacy ordering).

measured at the state-wave level (the FINRA NFCS fields the Big-Five questions every three years, five wave values over 2009 to 2023), while anomaly returns are at the firm-month level; bridging the scales requires the identifying variation to be cross-firm *within* state-month cells, not cross-state (which would conflate literacy with other state-level return determinants) and not cross-time (which would conflate it with aggregate dynamics). Second, the

channel transmits through local-investor community ties that take time to form and decay, so for firms that relocate headquarters the effective local base is in transition; a clean design restricts to stable-headquarters firms and reports the static versus point-in-time correction as a falsification check. Third, the prediction concerns how the moderation varies with ownership structure, so the design stratifies by within-stock institutional-ownership tercile rather than collapsing across IO levels.

### 3.2 The baseline specification

The headline specification implements these requirements as a two-way fixed-effects panel regression:

$$\begin{aligned}
r_{i,s,t} = & \alpha_s + \tau_t + \gamma (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy\_z}_{s(i,t),t}) \\
& + \delta_1 (\text{mom}_{i,t} \cdot \text{IV}_{i,t}) + \delta_2 (\text{mom}_{i,t} \cdot \text{literacy\_z}_{s(i,t),t}) + \delta_3 (\text{IV}_{i,t} \cdot \text{literacy\_z}_{s(i,t),t}) \\
& + \beta' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}.
\end{aligned} \tag{1}$$

The dependent variable  $r_{i,s,t}$  is the month- $t$  return of firm  $i$  headquartered in state  $s(i, t)$ . State fixed effects  $\alpha_s$  absorb time-invariant state characteristics; month fixed effects  $\tau_t$  absorb aggregate shocks. The three two-way interactions enter as controls; the three-way coefficient  $\gamma$  is the object of interest. The state-assignment function  $s(i, t)$  is the static Compustat snapshot

under the headline merge and the point-in-time EDGAR 10-K-header reassignment under the falsification check; on the stable-headquarters subsample the two coincide. The discrete-tercile rendering of Equation (1) re-estimates  $\gamma$  within each within-stock IO tercile.

### 3.3 Adding size as a moderator: why and how

Mid-IO firms are concentrated in mid-cap market-equity quintiles (median market capitalization roughly five to ten times higher than in low-IO), so a reader could ask whether the mid-IO three-way is a size moderation captured by the IO discretization. The size-controlled specification rules this out by entering size as a parallel moderator on the same anomaly:

$$\begin{aligned}
r_{i,s,t} = & \alpha_s + \tau_t + \gamma (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy\_z}_{s(i,t),t}) \\
& + \zeta (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{size\_z}_{i,t}) \\
& + \text{lower-order literacy interactions} + \text{lower-order size interactions} + \varepsilon_{i,s,t},
\end{aligned} \tag{2}$$

where  $\text{size\_z}_{i,t}$  is the within-month  $z$ -score of  $\log(\text{ME}_{i,t})$ , entered with the same standardization and interaction structure as literacy. The coefficient  $\gamma$  in Equation (2) is identified holding size moderation of the same anomaly fixed: if the literacy moderation is mechanically conflated with size through the mid-IO sample’s mid-cap concentration,  $\gamma$  should attenuate to insignifi-

cance; if literacy is a distinct moderator,  $\gamma$  should survive. Equation (2) is the preferred specification throughout.

### 3.4 Identifying assumptions

For  $\gamma$  to be interpretable as a literacy moderation, three assumptions must hold.

**Conditional unconfoundedness for the literacy moderator.** Conditional on state and month fixed effects, the three two-way interactions, and the size moderator in Equation (2), residual variation in state-aggregate literacy is uncorrelated with omitted return determinants that interact with  $\text{mom} \times \text{IV}$  in the same functional form. The paper addresses this four ways: a Fama-French 12-industry control block; a state college-attainment horse-race within mid-IO (literacy three-way  $-0.0350$ ,  $t = -2.97$ ; college  $+0.00076$ ,  $t = +0.65$ ); the within-stock IO contrast, which discriminates against any generic state-level moderator hitting all headquarters-state stocks alike; and the size-controlled specification, which discriminates against the most natural firm-level moderator. What remains is a class of ownership-interactive state-level confounds the design does not separate, which Section 7 takes up.

**Correct headquarters-state assignment.** The static Compustat snapshot mis-assigns relocater firm-months for the 14 percent of observations whose static state differs from the then-current 10-K headquarters state.

Headline patterns are estimated on the stable-HQ subsample, for which the static and point-in-time merges coincide; the PIT reassignment on the full panel is the strictest available correction, and the resulting magnitude collapse (Section 6) is the binding population-scope limitation.

**Cluster-robust inference with 51 state clusters.** The natural clustering dimension is the state, and with 51 clusters asymptotic cluster-robust standard errors over-reject; the binding procedure is the one-way state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed.

### 3.5 Causal language

Under conditional unconfoundedness,  $\hat{\gamma}$  in Equation (2) recovers the average within-state-month differential sensitivity of returns to  $\text{mom} \times \text{IV}$  as a function of state-aggregate literacy; it is the conditional association the design supports, not a clean average treatment effect.<sup>2</sup>

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<sup>2</sup>A natural instrument for state literacy is state K-12 personal-finance graduation requirements (building on Urban-Schmeiser 2015/2020), but it is not usable here: mandate states are systematically less literate, since states adopted mandates in response to low adult literacy, so the cross-state first stage is wrong-signed, and a cohort-matched version with state, wave, and age-band fixed effects yields  $F = 0.05$ . The OLS estimate is the preferred inferential statement. Detail is in IA V and Section 6.

## 4 Results

The introduction asked where in the cross-section the literacy mechanism becomes visible. This section answers in four steps: the moderation concentrates in mid-IO firms and survives size as a competing moderator; the finding is a property not of mid-IO firms specifically but of an underlying economic population whose IO location shifts with the sample’s size composition (the central mechanism evidence); that same population carries a cross-state literacy gradient in the predicted direction while firms outside it show none; and a parametric U-shape and a sample-selection decomposition close the section.

### 4.1 Where the moderation lives: a U-shape in institutional ownership

To see whether literacy moderates the momentum-IV interaction differently across firms, I discretize the within-stock IO distribution into terciles and estimate Equation (1) within each on the stable-HQ subsample. The naive expectation is that the channel is strongest where retail dominates (low IO) and fades where institutions dominate (high IO), with the middle tercile ambiguous. Table 2 reports the result under persistent and time-varying IO measures.

The naive prediction is rejected. The low-IO tercile, the most retail-driven and supposedly most literacy-sensitive, returns a null coefficient, and

Table 2: Mid-IO concentration: three-way coefficient by IO tercile, stable-HQ subsample

| <i>Panel A: Persistent IO measure</i>   |               |                |                     |              |         |
|-----------------------------------------|---------------|----------------|---------------------|--------------|---------|
| Tercile                                 | Mean IO share | $\hat{\gamma}$ | State-clustered $t$ | wcb $p$      | $N$     |
| IO1 (low; high retail)                  | 0.170         | −0.0113        | −1.04               | 0.296        | 144,989 |
| <b>IO2 (mid)</b>                        | <b>0.495</b>  | <b>−0.0321</b> | <b>−2.63</b>        | <b>0.064</b> | 191,589 |
| IO3 (high; institutional)               | 0.704         | −0.0050        | −0.31               | 0.760        | 182,384 |
| <i>Panel B: Time-varying IO measure</i> |               |                |                     |              |         |
| Tercile                                 | Mean IO share | $\hat{\gamma}$ | State-clustered $t$ | wcb $p$      | $N$     |
| IO1 (low; high retail)                  | 0.174         | −0.0087        | −0.92               | 0.337        | 141,129 |
| <b>IO2 (mid)</b>                        | <b>0.504</b>  | <b>−0.0265</b> | <b>−2.28</b>        | <b>0.058</b> | 184,881 |
| IO3 (high; institutional)               | 0.711         | −0.0044        | −0.25               | 0.789        | 167,879 |

*Notes.* Each row reports the three-way coefficient  $\hat{\gamma}$  on  $\text{mom} \times \text{IV} \times \text{literacy\_z}$  estimated within the IO tercile, on the stable-HQ subsample under static-snapshot literacy. The persistent measure is each firm’s time-mean institutional-ownership share; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier. Wild-cluster bootstrap uses Webb six-point weights [Cameron et al., 2008] with  $B = 4,999$  and null imposed. Sample: January 2009 through December 2023.

the high-IO tercile is also null, consistent with institutional dominance neutralizing the local-state channel. The moderation concentrates in the middle: mid-IO is the only tercile with state-clustered  $|t| > 2$  on both IO measures, with magnitude roughly three times the low-IO and six times the high-IO estimates. This discrete contrast does not depend on a functional form for IO and is the central characterization; Section 4.5 reports a continuous quadratic specification that corroborates the U-shape but does not replace it.

Table 3: Size-controlled mid-IO three-way: literacy survives

| IO tercile                                                                                        | Baseline $\hat{\gamma}$ (state- $t$ ) | Size-controlled $\hat{\gamma}$ (state- $t$ ) | $N$     |
|---------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------|---------|
| IO1 (low)                                                                                         | -0.0113 (-1.03)                       | -0.0077 (-0.65)                              | 144,989 |
| <b>IO2 (mid)</b>                                                                                  | <b>-0.0321 (-2.61)</b>                | <b>-0.0256 (-2.24)</b>                       | 191,589 |
| IO3 (high)                                                                                        | -0.0050 (-0.31)                       | -0.0001 (-0.00)                              | 182,384 |
| Horse race in mid-IO: $\gamma_{\text{size}}$ on $\text{mom} \cdot \text{IV} \cdot \text{size\_z}$ | (n/a)                                 | +0.0017 (+0.64)                              | 191,589 |

*Notes.* Baseline column re-estimates Equation (1) within IO tercile. Size-controlled column estimates Equation (2) within IO tercile, adding **size\_z**,  $\text{mom} \cdot \text{size\_z}$ ,  $\text{IV} \cdot \text{size\_z}$ , and  $\text{mom} \cdot \text{IV} \cdot \text{size\_z}$ . State and month fixed effects throughout. State-clustered  $t$  in parentheses. The horse-race row reports the coefficient on size's three-way under the size-controlled specification within mid-IO.

## 4.2 Is the U-shape just a size effect?

Mid-IO firms are also concentrated in the middle of the market-capitalization distribution (median ME \$561 million, versus \$66 million for low-IO and \$1.40 billion for high-IO), so a reader could wonder whether the finding is a size effect captured by the IO discretization. The size-controlled specification (Equation 2) tests this by entering size as a parallel moderator on the same anomaly, with the same standardization and interaction structure as literacy: if the mid-IO three-way is mechanically conflated with size, the literacy coefficient should attenuate to insignificance; if literacy is distinct, it should survive. Table 3 reports the comparison.

Literacy survives. The mid-IO three-way attenuates about twenty percent (to -0.0256) but stays significant at  $t = -2.24$ , while size's own three-way within mid-IO is statistically zero: conditional on being inside the mid-IO tercile, the literacy moderation is identified independently of size. The specification also sharpens the U-shape, turning the already-small high-IO

coefficient into a clean null ( $-0.0001$ ,  $t = -0.00$ ).

### 4.3 The migration test: what moves when the sample changes

The size-controlled result tells us the moderation lives in mid-IO but not *why*. A natural reading is that mid-IO is simply where the full panel’s size composition puts the interaction zone, so the moderation should follow that zone as the sample changes rather than being a fixed property of the tercile. Restricting to larger firms removes the micro-caps that populate low-IO and shifts the conditional IO distribution upward; under the interaction-zone reading, firms that occupied mid-IO at the full-panel margin should reappear at low-IO in the restricted sample, carrying the moderation with them.

Table 4 tests this: Panel A reports the within-panel three-way coefficient under progressive size filters, and Panel B the cross-state literacy gradient in the relevant cell under each filter.

The prediction holds cell by cell. In the full panel the moderation lives in mid-IO; dropping the bottom twenty percent of firms by market cap *strengthens* the mid-IO coefficient (the micro-caps removed fall predominantly in low-IO); and dropping the bottom fifty percent relocates the significant coefficient entirely to low-IO ( $\hat{\gamma}_{IO1} = -0.0441$ ,  $t = -2.04$ ), with mid-IO now null. The within-panel finding is not a property of mid-IO firms but of an underlying population whose IO label depends on what else is in the sam-

Table 4: Interaction-zone migration: IO tercile three-way coefficient under size filters, size-controlled spec

*Panel A: Within-panel three-way coefficient  $\hat{\gamma}_{lit}$  by IO tercile*

| Size filter                           | IO1 (low)              | IO2 (mid)              | IO3 (high)      | $N$     |
|---------------------------------------|------------------------|------------------------|-----------------|---------|
| Full stable-HQ panel (paper headline) | -0.0077 (-0.65)        | <b>-0.0256 (-2.24)</b> | -0.0001 (-0.00) | 518,962 |
| Drop bottom 20% (micro-cap)           | -0.0137 (-1.14)        | <b>-0.0402 (-2.57)</b> | +0.0109 (+0.75) | 454,107 |
| Drop bottom 33% (small-cap)           | -0.0155 (-1.02)        | -0.0273 (-1.29)        | +0.0122 (+0.66) | 395,873 |
| Drop bottom 50% (below median)        | <b>-0.0441 (-2.04)</b> | -0.0098 (-0.58)        | +0.0115 (+0.58) | 322,137 |

*Panel B: Cross-state weighted slope of per-state  $\hat{\gamma}_{lit}$  on state literacy*

| Sample cell                                   | Weighted slope | state- $t$   | Pearson $r$   | $n_{states}$ |
|-----------------------------------------------|----------------|--------------|---------------|--------------|
| Full panel, mid-IO (headline cell)            | <b>-0.0623</b> | <b>-5.01</b> | -0.275        | 42           |
| Drop bottom 50%, mid-IO                       | -0.0325        | -3.82        | +0.008        | 31           |
| Drop bottom 50%, IO1 (migrated headline cell) | <b>-0.0379</b> | <b>-4.30</b> | <b>-0.291</b> | 34           |
| Drop bottom 50%, IO3 (predicted null)         | +0.0061        | +0.26        | +0.054        | 31           |

*Notes.* Panel A: Equation (2) within each IO tercile, under each size filter applied at the firm level using firm panel-mean market capitalization. State and month fixed effects throughout. State-clustered  $t$  in parentheses. Bold cells indicate the tercile carrying the literacy moderation under each filter. Panel B: per-state  $\hat{\gamma}_{lit}$  from Equation (2) with month fixed effects only (state fixed effects collapse for single-state regressions); the cross-state regression of  $\hat{\gamma}_{lit}$  on state NFCS literacy uses precision weighting ( $1/SE^2$ ). States with fewer than 5 firms in the sample cell are excluded. The cross-state state- $t$  reported is the precision-weighted-slope  $t$ -statistic.

ple. The cross-state gradient migrates with it: it lives in mid-IO in the full panel ( $r = -0.28$ ,  $t = -5.01$ ) and in low-IO in the above-median sample ( $r = -0.29$ ,  $t = -4.30$ ), while high-IO shows none under either composition, the null the reading predicts. The migration is the signature an interaction-zone microfoundation predicts and the IO-tercile result alone does not deliver.

#### 4.4 Across states, does the moderation track literacy?

A direct cross-sectional test asks whether, across firms in the interaction-zone cell, the moderation tracks state literacy in the predicted direction. I

estimate the three-way coefficient state by state under Equation (2) restricted to that cell, then regress the per-state coefficients on state NFCS literacy with precision weighting:

$$\hat{\gamma}_{\text{lit},s} = \alpha + \beta \cdot \text{literacy}_s + u_s, \quad \hat{\beta} = -0.0623, \text{ SE} = 0.0124, t = -5.01, \text{ Pearson } r = -0.275. \quad (3)$$

The gradient runs in the predicted direction: higher-literacy states show more negative per-state coefficients, with a state-permutation two-sided  $p = 0.0047$ . Across the three jointly relevant tests, the headline coefficient, the high-IO coefficient, and the cross-state slope, Holm step-down family-wise  $p$ -values are 0.05, 0.014, and 0.996 respectively: the headline and the gradient survive multiplicity correction, and the high-IO cell is empty.

The same negative slope appears in two settings, the full-panel mid-IO cell and the above-median IO1 cell, recovering essentially the same gradient across two IO terciles in two sample restrictions (IA Y), as the migration story predicts. Figure 1 consolidates the migration evidence: the within-panel coefficients across the four size filters, and the cross-state gradient scatters for the original headline cell, the cell that loses the gradient when smaller firms are removed, and the migrated headline cell. The interaction zone moves with the sample, and the gradient moves with it.

**Figure 2: Interaction-zone migration across sample size restrictions**  
**The mid-IO finding is sample-conditional. The underlying object is the size  $\times$  ownership interaction zone.**

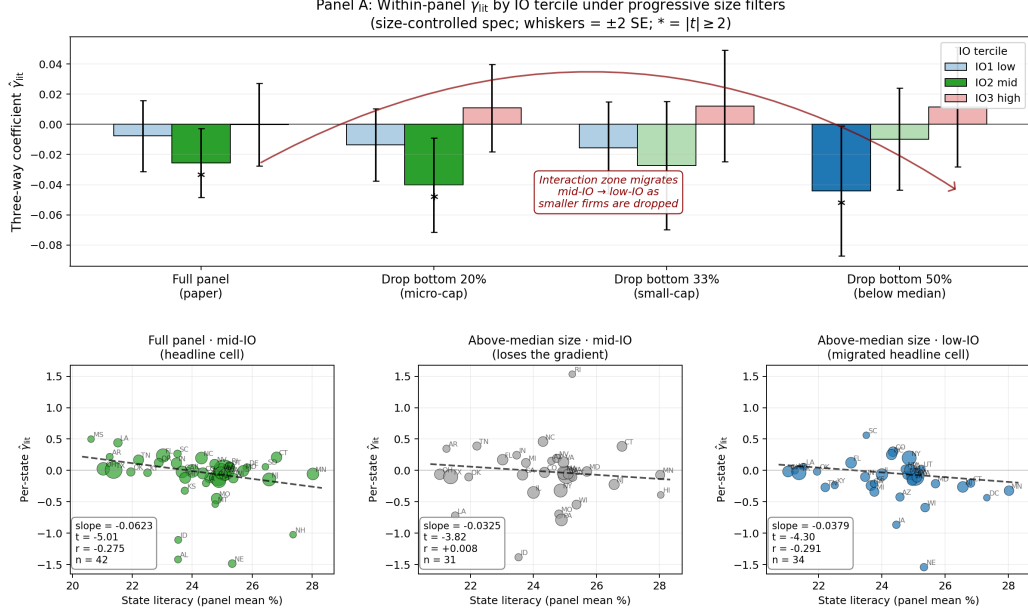


Figure 1: Interaction-zone migration across sample size restrictions. Panel A: within-panel  $\hat{\gamma}_{lit}$  by IO tercile under progressive size filters, size-controlled spec, whiskers =  $\pm 2$  SE,  $\star = |t| \geq 2$ . Panel B: cross-state scatter of per-state  $\hat{\gamma}_{lit}$  on state literacy for three sample cells. Point size  $\propto \sqrt{n_{\text{firms}}}$ .

## 4.5 A continuous corroboration of the U-shape

The discrete-tercile contrast is the headline result. A natural follow-up asks whether a continuous functional form on IO recovers the same non-monotone shape with a separable interior trough. Augmenting Equation (1) with linear and quadratic IO interactions produces the test:

$$r_{i,s,t} = \alpha_s + \tau_t + \gamma_4 (\text{mom}\cdot\text{IV}\cdot\text{literacy\_z}\cdot\text{IO}) + \gamma_5 (\text{mom}\cdot\text{IV}\cdot\text{literacy\_z}\cdot\text{IO}^2) + \beta' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}, \quad (4)$$

where  $\mathbf{X}$  collects the 21 lower-order interactions and the literacy slope  $s(\text{IO}) = \gamma_3 + \gamma_4 \text{IO} + \gamma_5 \text{IO}^2$  has an interior minimum at  $\text{IO}^* = -\gamma_4/(2\gamma_5)$  when  $\gamma_5 > 0$ . On the stable-HQ persistent measure  $\hat{\gamma}_5 = +0.0151$  ( $t = +2.28$ , wcb  $p = 0.017$ ), with implied trough  $\text{IO}^* = 0.344$ , inside the mid-IO tercile and where the discrete contrast predicts. The clearance rests on a single functional form, since log-IO and kink-IO specifications return insignificant curvature (Section 6), so the discrete-tercile contrast remains the robust claim and the quadratic is corroboration rather than a separate inferential statement.

## 4.6 How much of the contrast comes from sample selection?

The headline drops relocators, so part of the mid-vs-low contrast reflects the firms excluded. The mid/low coefficient ratio is 2.83 on the stable-HQ panel against 1.09 on the full panel including relocators. A firm-by-firm decomposition (IA G) attributes about one-third of the gap to the low-IO comparator weakening under the relocator drop (relocators are nearly twice as concentrated in low-IO, with a near-zero relocator-only low-IO coefficient) and about two-thirds to mid-IO strengthening. Relocators are smaller, higher-IV,

lower-IO, lower-return firms, so part of the headline magnitude reflects sample composition. Section 6 takes up the population-scope question directly through the point-in-time literacy reassignment on the full panel.

## 5 Mechanism: The Interaction-Zone Characterization

Section 4 documented four facts: the moderation concentrates in mid-IO ( $\hat{\gamma}_{\text{mid}} = -0.0256$ ,  $t = -2.24$ ); it relocates to low-IO when the bottom 50 percent of firms by market cap are dropped ( $\hat{\gamma}_{\text{IO1}} = -0.0441$ ,  $t = -2.04$ ); the per-state coefficient tracks state literacy in the cell that carries the moderation ( $r = -0.28$  full-panel mid-IO,  $r = -0.29$  above-median IO1) but not in high-IO ( $r = -0.07$ ); and a decomposition by idiosyncratic-volatility decile within mid-IO shows no concentration at high IV. This section develops the interaction-zone characterization those facts identify and notes what the design does and does not separate.

### 5.1 The interaction-zone reading

The moderation lives where two conditions hold at once: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into cross-state return differences. Where neither holds (micro-caps held mostly by retail), mispricing accumulates but does not load on cross-

state literacy variation; where institutions dominate (large high-IO firms), arbitrage absorbs retail demand and the marginal investor is not a local trader. The interaction zone is the set of mid-cap firms with mixed ownership, where both conditions hold. Its IO label is sample-conditional: under the full-panel size composition it falls in mid-IO, because mid-IO firms are predominantly mid-cap, and under the above-median composition it migrates to low-IO, because low IO among large firms means retail-tilted at substantial size. Table 4 traces this cell by cell, and the high-IO null under both compositions is the null the reading predicts.

## 5.2 Not a limits-to-arbitrage effect

The textbook limits-to-arbitrage microfoundation [Shleifer and Vishny, 1997, Pontiff, 2006] predicts that literacy-modulated retail demand should displace more mispricing where arbitrage is costlier, that is, at higher idiosyncratic volatility. A within-month NYSE IV-decile decomposition within mid-IO does not show this: the highest-IV decile is insignificant ( $\hat{\gamma} = -0.035$ ,  $t = -1.27$ ) and a joint equality test across deciles does not reject ( $\chi^2(9)$ ,  $p = 0.94$ ), with the full table in IA X. The channel I identify therefore operates through size and ownership conditioning, pinned down by the migration of Section 4.3, rather than through arbitrage cost along the IV margin.

### 5.3 High-IO null and college horse-race

The high-IO cross-state null ( $\hat{\beta}_{\text{IO3}} = +0.018$ ,  $t = +1.22$ ,  $r = -0.07$ ; Holm  $p_{\text{FWER}} = 0.996$ ) is the second pattern the reading anticipates: high-IO firms are held by nationally diversified institutions indifferent to headquarters state, so state-level literacy variation says nothing about the marginal investor, and whatever retail demand exists is arbitrated away. Two diagnostics rule out the obvious alternatives. Literacy is not a proxy for education: a state college-attainment horse-race within mid-IO leaves literacy sharp ( $\hat{\gamma}_{\text{lit}} = -0.0350$ ,  $t = -2.97$ ) and college null ( $+0.00076$ ,  $t = +0.65$ ) despite a  $\rho = 0.62$  state-level correlation. And the mid-IO concentration is not a variance artifact, since mid-IO has lower identifying variation than low-IO yet a larger slope (Section 6).

### 5.4 What the design does not separate

The characterization is identified through the size-restriction migration and the cross-state gradient, but it does not separate three readings that share the same size and ownership conditioning: literacy-modulated retail demand *intensity* (the reading I most naturally adopt, consistent with the gradient direction), literacy-modulated retail *attention* (less momentum-following pressure in high-literacy states), and literacy-modulated retail *timing* (a clientele-timing channel generating the same gradient). The three make different predictions for retail trade-flow data (volume, turnover, cross-spectral sig-

natures) not available in the current panel; the TAQ-based Boehmer-Jones-Zhang retail trade indicator would separate them, a natural next-paper extension.

## 5.5 Population scope

The characterization holds on the stable-HQ subsample, on which the local-investor unit of analysis is held fixed. Under the point-in-time 10-K-header reassignment the mid-IO coefficient shrinks sharply (Section 6): the directional within-tercile pattern survives but the magnitude does not, so the characterization is a statement about stable-headquarters firms rather than the corrected full panel.

## 6 Robustness

This section reports the robustness checks. The point-in-time (PIT) headquarters correction bounds the population scope of the result and is the binding check, so I report it in full; the remaining checks are summarized here, with detail in the Internet Appendix.

Table 5: The nested baseline: aggregate and mid-IO three-way under PIT correction

| <i>Panel A: Aggregate three-way coefficient</i> |                |            |         |         |
|-------------------------------------------------|----------------|------------|---------|---------|
| Specification                                   | $\hat{\gamma}$ | State- $t$ | wcb $p$ | $N$     |
| Full panel, static-snapshot literacy            | −0.0117        | −2.17      | 0.078   | 623,896 |
| Full panel, PIT literacy reassignment           | −0.0003        | −0.86      | 0.517   | 623,814 |
| Stable-HQ subsample, static literacy            | −0.0145        | −1.94      | 0.122   | 519,090 |

| <i>Panel B: Mid-IO coefficient under PIT, with and without size controls</i> |                                |            |                            |         |
|------------------------------------------------------------------------------|--------------------------------|------------|----------------------------|---------|
| Specification                                                                | $\hat{\gamma}_{\text{mid-IO}}$ | State- $t$ | Collapse vs. baseline      | $N$     |
| Stable-HQ static, persistent (paper headline)                                | −0.0321                        | −2.63      | (baseline)                 | 191,589 |
| Stable-HQ static, size-controlled                                            | −0.0256                        | −2.24      | (baseline)                 | 191,589 |
| PIT full panel, persistent                                                   | −0.00121                       | −2.07      | $\approx 27\times$ smaller | 191,589 |
| PIT full panel, size-controlled                                              | −0.00100                       | −1.75      | $\approx 26\times$ smaller | 191,589 |

*Notes.* Static-snapshot merge pairs each firm-month with the literacy of its Compustat **state**; PIT reassignment re-merges against the firm’s then-current 10-K-header state and re- $z$ -scores within month. State-level wild-cluster bootstrap (Webb six-point,  $B = 4,999$  or  $9,999$ , null imposed). The PIT magnitude collapse is essentially the same with and without size controls ( $27\times$  vs.  $26\times$ ).

## 6.1 The PIT correction and the nested-baseline collapse

The seed prior to this paper’s revisions predicted a negative aggregate three-way on the full panel and a negative within-stock retail-versus-institutional difference. Table 5 reports both objects under the static-snapshot literacy merge and under the point-in-time EDGAR 10-K-header literacy reassignment.

Two readings of the collapse are possible. A mechanism-strength reading says the channel operates most strongly on stable-HQ firms, where the local-investor unit of analysis is well-defined, so adding back the relocated

firm-months dilutes the average. A sample-selection reading says the stable-HQ subsample is tilted toward firms with stronger anomaly moderation for reasons unrelated to literacy. The relocater characteristics in IA G (relocators are smaller, higher-IV, lower-IO, lower-return firms) favor the sample-selection reading. Either way, the characterization is a statement about the stable-HQ population, and generalizing to the corrected full panel is the main population-scope limitation.

## 6.2 Time variation in the cross-state gradient

The within-panel mid-IO three-way coefficient is essentially identical pre- and post-2015 ( $\hat{\gamma}^{\text{pre}} = -0.024$ ,  $\hat{\gamma}^{\text{post}} = -0.026$ ; difference  $-0.002$ ,  $t = +0.06$ ): the within-panel direction is stable. What varies across time is the *cross-state* literacy gradient that interprets the result through the literacy mechanism. The pooled gradient is right-signed and sharp ( $-0.0623$ ,  $\text{state-}t = -5.01$ ), and the post-2015 sub-period is sharper still ( $-0.0706$ ,  $t = -3.50$ ); the pre-2015 sub-period is wrong-signed ( $+0.0403$ ,  $t = +1.89$ ) and concentrated in NFCS Wave 2009, the first administration of the Big-Five module (Waves 2012 to 2021 are directionally negative). The most likely explanation is measurement maturation in the later NFCS waves rather than a 2015-era regime change: institutional ownership of the panel rose post-2015 (mean  $\Delta \text{io\_share} = +0.083$ ), which rules out a retail-trading regime shift, and the state-level change in IO is uncorrelated with state literacy ( $r = +0.04$ ). The sub-period table and a time-variation figure are in IA W and IA Y.

### 6.3 Additional checks

The remaining checks are reported in full in the Internet Appendix and summarized here.

*Functional form.* The continuous-margin quadratic-IO clearance ( $\hat{\gamma}_5 = +0.0151$ ,  $p = 0.017$ ) does not survive log-IO ( $p = 0.75$ ), kink-IO ( $p = 0.24$ ), or a flat twenty-bin diagnostic, so the discrete-tercile contrast of Table 2 carries the claim. The grid is IA U.

*Instrumental variables.* The K-12 personal-finance mandate instrument is not usable: the strongest candidate has  $r = -0.355$  (wrong sign) and  $F = 7.06$ , because low-literacy regions adopted mandates as a remedy, and a cohort-matched version with state, wave, and age-band fixed effects returns  $F = 0.05$ . OLS from Equation (2) is the preferred inferential statement, with first-stage and 2SLS detail in IA V.

*Variance channel and education.* The mid-IO concentration is not a variance artifact, since mid-IO has lower identifying variation than low-IO yet a larger slope, and a state college-attainment horse-race within mid-IO leaves literacy sharp ( $-0.0350$ ,  $t = -2.97$ ) and college null ( $+0.00076$ ,  $t = +0.65$ ), so literacy is not a proxy for education. A within-retail literacy split inside the low-IO tercile adds nothing once decomposed across IV deciles (IA C).

*Inference, wave-anchor, and vintage.* One-way state clustering on 51 clusters via the wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed is the binding procedure

throughout.<sup>3</sup> Restricting to the five NFCS wave-anchor years sharpens the mid-IO coefficient ( $-0.070$ ,  $t = -2.09$ ), so the headline is not a forward-fill artifact, and fixing each firm’s IO tercile to its 2009 assignment attenuates the mid/low ratio from 2.83 to 2.09 but preserves the concentration.

## 7 Discussion

### 7.1 What the characterization changes

The characterization positions state NFCS Big-Five literacy as a cross-sectional asset-pricing moderator that operates through an interaction-zone microfoundation, identified through the size and ownership margin rather than the IV margin the canonical limits-to-arbitrage reading predicts. Where household finance has studied literacy as a predictor of portfolio quality, participation, and resilience [Lusardi and Mitchell, 2014, Calvet et al., 2007, Grinblatt et al., 2012], this paper makes state-aggregate literacy a moderator on a specific anomaly interaction with a precisely characterized population scope. It adds a literacy moderator to the local-return-predictability program of Korniotis and Kumar [2013]: the two are distinct, since literacy is identified through IO and size conditioning the business cycle does not predict, but partly overlapping, since a business-cycle control block attenuates the literacy three-way by 20 to 23 percent.

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<sup>3</sup>The cross-state gradient is interpreted alongside a state-permutation  $p$ -value ( $B = 9,999$ ), with Holm step-down family-wise correction across the three jointly relevant tests. The two-way state-by-month CGM statistic is reported as a diagnostic in IA I.

Taking the size-controlled mid-IO coefficient  $\hat{\gamma}_{\text{mid-IO}} = -0.0256$ , the economic magnitude is a literacy-modulated annualized return differential of about  $-5.5\%$  at a 0.187-standard-deviation swing in the standardized composite. The non-size-controlled coefficient implies  $-7\%$ , and a classical errors-in-variables correction (reliability ratio  $\lambda = 0.420$ ) extends the bracket to roughly  $-13\%$ . This  $-5.5\%$  to  $-13\%$  bracket overlaps the  $\approx 5\%$  state-business-cycle strategy of Korniotis and Kumar [2013].

## 7.2 Limitations

**Population scope.** On the PIT-corrected full panel the mid-IO coefficient shrinks sharply, and the relocater characteristics (IA G) point toward a sample-selection rather than mechanism-strength reading of that collapse: relocators are smaller, higher-IV, lower-IO, lower-return firms, and about one-third of the stable-HQ mid/low gap comes from the low-IO comparator weakening under the relocater drop. The characterization is a statement about firms with stable headquarters across the panel window. Pre and post-relocation event studies on the 1,366 dropped firms would discriminate between the two readings.

**Microfoundation and time variation.** The characterization is identified through the size-restriction migration and the cross-state gradient but does not separate the within-family readings (demand intensity, attention, timing), which the Boehmer-Jones-Zhang retail trade indicator could distin-

guish. The pooled cross-state gradient is driven mainly by post-2015 data; the pre-2015 slope is wrong-signed and concentrated in NFCS Wave 2009. Since institutional ownership rose post-2015 and is uncorrelated with the literacy gradient, and the within-panel mid-IO coefficient is essentially identical pre and post-2015, this looks like measurement noise in the first Big-Five administration rather than a substantive shift.

**Identification.** The K-12 personal-finance mandate instrument is endogenous to state literacy through reverse causality, so the OLS estimate is the preferred inferential statement; a clean instrument would need pre-1990 mandate variation [Bernheim et al., 2001] or verifiable-compliance cohort micro-data. The within-stock contrast also does not separate an ownership-interactive state-level confound, the leading instance being state-resident public-pension home-state ownership (about 21 managers in roughly seven states, collinear with the state fixed effect elsewhere), which would require a manager-level state-pension flag to close.

## 8 Conclusion

A large literature documents that less financially literate investors trade systematically worse, and a separate literature documents that retail demand is geographically concentrated and moves prices. Together they predict that state-level investor literacy should moderate the return anomalies that de-

pend on retail mispricing. Aggregate tests deliver weak signals; the contribution of this paper is to show that the literacy mechanism is not absent but hidden by aggregation, and to characterize where in the cross-section, and under what conditions on ownership and size, it operates.

The data identify an interaction zone. The literacy channel leaves a price footprint only where retail demand is large enough to move prices and institutional activity is sufficient to translate it into return differences across states. Where neither holds (micro-caps held mostly by retail) mispricing accumulates without loading on state literacy; where institutions dominate (large high-IO firms) arbitrage neutralizes the local-state channel. In the full stable-HQ panel the zone falls in mid-IO; restricting to larger firms moves it to low-IO, and the cross-state literacy gradient moves with it, while firms outside the zone show no gradient. This migration with sample composition is the paper’s central piece of evidence, and it identifies the interaction-zone reading without relying on an instrument. The result rests on OLS under conditional unconfoundedness, with the migration supplying the predicted-signature confirmation an instrument would otherwise provide.

The characterization holds for firms with stable headquarters; under the point-in-time literacy reassignment the headline magnitude shrinks, which bounds the population to which it applies, and the U-shape across the IO distribution is robust as a discrete-tercile contrast rather than as a parametric form. Whether other state-level investor traits such as participation rates, brokerage density, or retail-attention measures operate through a similar in-

teraction zone is a natural next question, and the size-restriction migration test developed here gives a portable way to look.

## A Variable construction detail

Extended robustness material is in the Internet Appendix: the Herfindahl-Hirschman alternative (IA M), the Korniotis-Kumar business-cycle block (IA O), the difference-in-differences secondary design (IA T), the full PIT literacy reassignment (IA A), the relocater-vs-stable-HQ characteristics comparison (IA G), the Thomson Reuters s34 panel construction (IA P), and alternative literacy constructions (IA L).

**NFCS Big Five share-correct score.** The state financial-literacy measure is the share of NFCS respondents in a state-wave who answer the FINRA Big Five literacy questions correctly. Correct-answer keys from the FINRA codebook: M6=1, M7=3, M8=2, M9=2, M10=1. An earlier build coded all five questions with a single key; the corrected keys are used throughout, and the Table 1 state-literacy rankings are post-correction. State-wave scores are forward-filled to monthly frequency between waves, then z-scored within calendar month to form `literacy_z`.

**Idiosyncratic volatility.** Standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window ending in the observation month; z-scored within calendar month before entering the interaction

terms.

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